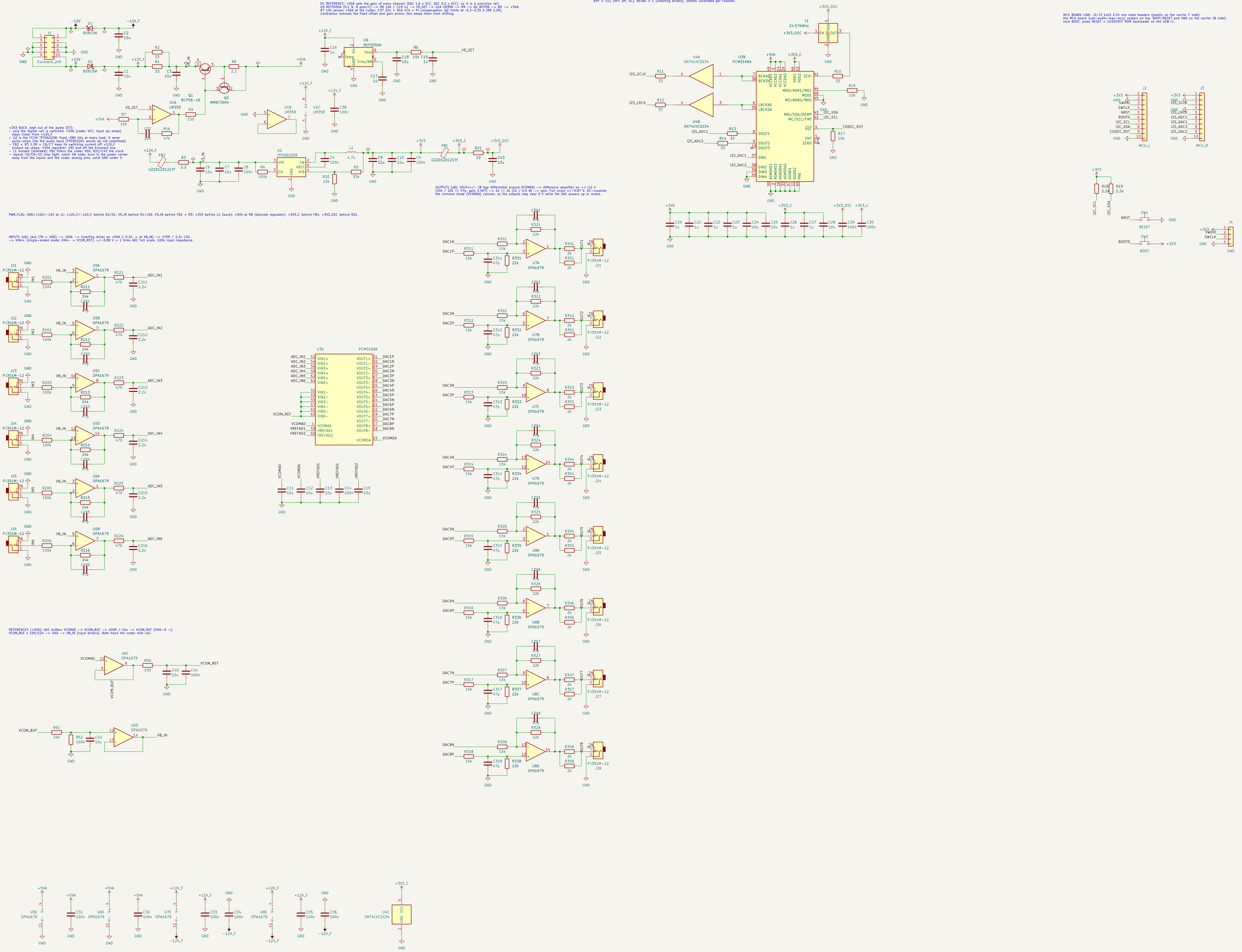
BIG MODULAR USB–AUDIO–MAX – carrier (jack board), S2. 6 in / 8 out, DC–coupled +/–5.9 V, no level controls, PCM3168A 24–bit, up to 96 kHz. Circuit intent: docs/research.md

POWER: +12V –> D2 –> +12V_F (output op–amps; R1||R2 16.5R –> Q1 BCP56 (U1A LM358 loop, ref U9 REF5050) –> +5VA; FB2 + R3 –> U2 TPS562208 FCCM buck –> L1 –> +3V3 –> FB1 –> +3V3_C –> R21 –> +3V3_OSC); –12V –> D1 –> –12V_F. +12 V ~0.34 A max, –12 V ~20 mA.



DC REFERENCE: +5VA sets the gain of every channel (DAC 1.6 x VCC, ADC 0.2 x VCC), so it is a precision rail:
U9 REF5050A (0.1 %, 8 ppm/C) –> R6 10k / C19 1u –> V5_SET –> U1A LM358 –> R9 –> Q1 BCP56 –> R8 –> +5VA;
R7 10k senses +5VA at the codec; C37 22n + R16 47k = PI compensation. Q2 limits at ~0.3–0.35 A (R8 2.2R).
Calibration removes the fixed offset and gain errors; this keeps them from drifting.

PCM3168A INTERFACE: I2C 0x44 (MODE = DGND, ADR1/ADR0 = 0). ADC = clock master, 24–bit I2S–mode TDM;
DAC = slave on the same BCK/LRCK (pins tied). SCKI 24.576 MHz: 96 kHz = 256 fS (dual rate, BCK 128 fS),
48 kHz = 512 fS (single rate, BCK 256 fS): BCK = 12.288 MHz in both. 96 kHz: DOUT1 ADC1–4, DOUT2 ADC5–6,
DIN1 DAC1–4, DIN2 DAC5–8. 48 kHz: DOUT1 ADC1–6, DIN1 DAC1–8. Firmware: SEAD = 1 (single–ended),
BYP = 111 (HPF off, DC), REVAD = 1 (inverting drivers), offsets calibrated per channel.

MCU BOARD LINK: J2/J3 1x10 2.54 mm male headers (plastic on the carrier F side);
the MCU board (usb–audio–max–mcu) solders on top. BOOT/RESET and SWD on the carrier (B side):
hold BOOT, press RESET = CH32V307 ROM bootloader on the USB–C.

+3V3 BUCK, kept out of the audio (D7):
- only the digital rail is switched; +5VA (codec VCC, input op–amps)
  stays linear from +12V_F
- U2 is the FCCM TPS562208: fixed ~580 kHz at every load, it never
  pulse–skips into the audio band (TPS562201 would: do not substitute)
- FB2 + R3 2.2R + C6/C7 keep its switching current off +12V_F
  (output op–amps, +5VA regulator, U9) and off the Eurorack bus
- L1 molded (shielded); FB1 filters the codec VDD, R21/C43 the clock
- layout: C6/C8–U2 loop tight, small SW node, buck in the power corner
  away from the inputs and the codec analog pins, solid GND under it

OUTPUTS (x8): VOUTn+/– (8 Vpp differential around VCOMDA) –> difference amplifier on +/–12 V
(15k / 22k || 47p, gain 1.467) –> 2k || 2k (1k / 0.5 W) –> jack. Full scale +/–5.87 V, DC–coupled:
the common mode (VCOMDA) cancels, so the outputs stay near 0 V while the DAC powers up or mutes.

PWR_FLAG: GND/+12V/–12V at J1; +12V_F/–12V_F behind D2/D1; V5_IN behind R1||R2; V3_IN behind FB2 + R3; +3V3 behind L1 (buck); +5VA at R8 (discrete regulator); +3V3_C behind FB1; +3V3_OSC behind R21.

INPUTS (x6): jack (TN = GND) –> 100k –> inverting driver on +5VA (–0.24, + at VB_IN) –> 470R / 2.2n C0G
–> VINn+ (single–ended mode; VINn– = VCOM_REF). +/–5.89 V = 1 Vrms ADC full scale. 100k input impedance.

REFERENCES (+5VA): U6C buffers VCOMAD –> VCOM_BUF –> 100R / 10u –> VCOM_REF (VIN1–6 –);
VCOM_BUF x 100/124 –> U6D –> VB_IN (input drivers). Both track the codec mid–rail.

Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S2 – 6 in / 8 out DC–coupled USB audio, PCM3168A codec (TDM, up to 96 kHz), +/–5.9
Big Modular
Sheet: /
File: usb–audio–max.kicad_sch
Title: Big Modular – USB–AUDIO–MAX carrier (jack board)
Size: A1
Date:
Rev: S2
KiCad E.D.A. 10.0.6
Id: 1/1